
DMFT Masked Hopfield

Carlo Lucibello
carlo.lucibello@unibocconi.it
Bocconi University, Milan

Filippo Elgorni
filippo.elgorni@unibocconi.it
Bocconi University, Milan

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ABSTRACT

1 Autoregressive Hopfield Setting

Consider $x_i = \pm 1$, $i = 1 \dots L$, and M patterns $\xi^\mu \sim \text{Unif}(\{-1, +1\}^L)$. Define $W_{ij} = \frac{1}{\sqrt{L}} \sum_\mu \xi_i^\mu \xi_j^\mu$.

The probability of a trajectory x in an autoregressive generation is given by

$$p_W(x) = \prod_i p_W(x_i \mid x_{<i}) = \prod_i \sigma \left(\beta x_i \sum_{j<i} \frac{W_{ij}}{\sqrt{L}} x_j \right) \quad (1)$$

where $\beta > 0$ is the inverse temperature controlling the sharpness of the conditional, and $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic. The generating functional of Martin-Siggia-Rose is given by

$$\mathbb{E}_W Z_W(\psi) = \mathbb{E}_W \sum_x p_W(x) e^{\sum_i \psi_i x_i} \quad (2)$$

Where the expectation over W is intended as the expectation over the patterns ξ^μ . We can write it as

$$\begin{aligned} \mathbb{E}_W Z_W(\psi) &= \mathbb{E}_W \int \prod_i \frac{dh_i d\hat{h}_i}{2\pi} \sum_x e^{\sum_i \psi_i x_i - i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i x_i \sum_{j<i} \frac{W_{ij}}{\sqrt{L}} x_j} \prod_i \sigma(\beta h_i) \\ &= \mathbb{E}_{\{\xi^\mu\}_{\mu=1}^M} \int \prod_i \frac{dh_i d\hat{h}_i}{2\pi} \sum_x e^{\sum_i \psi_i x_i - i \sum_i \hat{h}_i h_i + i \frac{1}{L} \sum_{\mu,i} \hat{h}_i x_i \sum_{j<i} \xi_i^\mu \xi_j^\mu x_j} \prod_i \sigma(\beta h_i) \end{aligned} \quad (3)$$

1.1 Partial-overlap order parameters

The pattern indices appear inside a triangular sum that is not a perfect square, so the Hubbard–Stratonovich trick of the symmetric Hopfield case (see appendix) cannot be applied directly. Following the MSR/DMFT logic we instead introduce, for each pattern $\mu = 1, \dots, M$ and each site i , the partial overlap

$$m_i^\mu := \frac{1}{L} \sum_{j<i} \xi_j^\mu x_j, \quad (4)$$

via a Fourier representation of the delta function:

$$1 = \int \prod_{\mu,i} \frac{L dm_i^\mu d\hat{m}_i^\mu}{2\pi} e^{iL \sum_{\mu,i} \hat{m}_i^\mu \left(m_i^\mu - \frac{1}{L} \sum_{j<i} \xi_j^\mu x_j \right)}. \quad (5)$$

The pattern-dependent piece of the action then reads

$$\frac{i}{L} \sum_{\mu,i} \hat{h}_i x_i \xi_i^\mu \sum_{j<i} \xi_j^\mu x_j = i \sum_{\mu,i} \hat{h}_i x_i \xi_i^\mu m_i^\mu, \quad (6)$$

and rearranging the auxiliary term by swapping the order of summation,

$$\sum_i \hat{m}_i^\mu \sum_{j < i} \xi_j^\mu x_j = \sum_j \xi_j^\mu x_j \hat{M}_j^\mu, \quad \hat{M}_i^\mu := \sum_{k > i} \hat{m}_k^\mu. \quad (7)$$

Each $\xi_i^\mu \in \{\pm 1\}$ now enters linearly, so the disorder average factorises across (μ, i) :

$$\mathbb{E}_{\xi_i^\mu} e^{i \xi_i^\mu x_i (\hat{h}_i m_i^\mu - \hat{M}_i^\mu)} = \cos(x_i (\hat{h}_i m_i^\mu - \hat{M}_i^\mu)) = \cos(\hat{h}_i m_i^\mu - \hat{M}_i^\mu), \quad (8)$$

where the last equality uses $x_i = \pm 1$ and the parity of $\cos$. Collecting all factors,

$$\begin{aligned} \mathbb{E}_W Z_W(\psi) &= \int [dh \, d\hat{h}] [dm \, d\hat{m}] \sum_x e^{\sum_i \psi_i x_i - i \sum_i \hat{h}_i h_i + iL \sum_{\mu,i} \hat{m}_i^\mu m_i^\mu} \prod_i \sigma(\beta h_i) \prod_{\mu,i} \cos(\hat{h}_i m_i^\mu - \hat{M}_i^\mu) \\ &= \int [dh \, d\hat{h}] [dm \, d\hat{m}] e^{\sum_i \log 2 \cosh(\psi_i) + \sum_i \log \sigma(\beta h_i) - i \sum_i \hat{h}_i h_i + \sum_{\mu,i} \log \cos(\hat{h}_i m_i^\mu - \hat{M}_i^\mu) + iL \sum_{\mu,i} \hat{m}_i^\mu m_i^\mu} \end{aligned} \quad (9)$$

with shorthand $[dh \, d\hat{h}] = \prod_i \frac{dh_i d\hat{h}_i}{2\pi}$ and $[dm \, d\hat{m}] = \prod_{\mu,i} \frac{L dm_i^\mu d\hat{m}_i^\mu}{2\pi}$.

1.2 Warmup: paramagnetic regime

We now make the **DMFT closure**: assume that, at the relevant saddle, the cosine arguments

$$\theta_i^\mu := \hat{h}_i m_i^\mu - \hat{M}_i^\mu \quad (10)$$

are small, so that

$$\log \cos(\theta_i^\mu) = -\frac{1}{2}(\theta_i^\mu)^2 + \sum_{k \geq 2} \kappa_{2k} \frac{\theta_i^{2k}}{(2k)!}, \quad (11)$$

and the higher cumulants κ_{2k} are neglected. This is a **closure assumption**, not an exact statement: while the second-cumulant calculation below is exact for finite-dimensional marginals of the disorder average, the full L -site generating functional generally retains non-Gaussian contributions that the present treatment discards. Note moreover that the bound $m_i^\mu = O\left(\frac{1}{\sqrt{L}}\right)$ alone does not control $\hat{M}_i^\mu = \sum_{k > i} \hat{m}_k^\mu$, which is a sum of $L - i$ small quantities; the small-cosine expansion is therefore an assumption on θ_i^μ itself, to be verified a posteriori at the saddle.

Under this closure,

$$\log \prod_{\mu,i} \cos(\theta_i^\mu(m_i^\mu, \hat{m}_i^\mu)) \approx -\frac{1}{2} \sum_{\mu,i} \left[\hat{h}_i^2 (m_i^\mu)^2 - 2\hat{h}_i m_i^\mu \hat{M}_i^\mu + (\hat{M}_i^\mu)^2 \right]. \quad (12)$$

The pattern indices factorise; for each μ the Gaussian integrals over $m_i^\mu, \hat{m}_i^\mu$ can be done explicitly.

$$\begin{aligned} &\int \frac{dm_i^\mu}{\sqrt{2\pi}} e^{-\frac{1}{2} [\hat{h}_i^2 (m_i^\mu)^2 - 2\hat{h}_i m_i^\mu \hat{M}_i^\mu + (\hat{M}_i^\mu)^2] + iL \hat{m}_i^\mu m_i^\mu} = \\ &= e^{-\frac{1}{2} (\hat{M}_i^\mu)^2} \int \frac{dm_i^\mu}{\sqrt{2\pi}} e^{-\frac{1}{2} \hat{h}_i^2 (m_i^\mu)^2 + m_i^\mu (\hat{h}_i \hat{M}_i^\mu + iL \hat{m}_i^\mu)} = \\ &= \frac{1}{\hat{h}_i} e^{-\frac{1}{2} \left[(\hat{M}_i^\mu)^2 - \frac{(\hat{h}_i \hat{M}_i^\mu + iL \hat{m}_i^\mu)^2}{\hat{h}_i^2} \right]} \\ &= \frac{1}{\hat{h}_i} e^{\frac{1}{2} \left[\frac{i2\hat{h}_i \hat{M}_i^\mu L \hat{m}_i^\mu - L^2 (\hat{m}_i^\mu)^2}{\hat{h}_i^2} \right]} \end{aligned} \quad (13)$$

so the next integral is:

$$\begin{aligned}
& \int \prod_i \frac{d\hat{m}_i^\mu}{\sqrt{2\pi}} e^{\frac{1}{2} \sum_i \left[\frac{i 2 \hat{h}_i L \hat{m}_i^\mu \sum_{k>i} \hat{m}_k^\mu - L^2 (\hat{m}_i^\mu)^2}{\hat{h}_i^2} \right]} = \\
& \int \prod_i \frac{d\hat{m}_i^\mu}{\sqrt{2\pi}} e^{i \frac{L}{\hat{h}_i} \hat{m}_i^\mu \sum_{i,k>i} \hat{m}_k^\mu - \frac{1}{2} \sum_i \left(\frac{L}{\hat{h}_i} \right)^2 (\hat{m}_i^\mu)^2} = \\
& \int \prod_i \frac{d\hat{m}_i^\mu}{\sqrt{2\pi}} e^{-\frac{1}{2} \sum_{ij} \hat{m}_i^\mu A_{ij} \hat{m}_j^\mu} = \frac{1}{\sqrt{\det A}}
\end{aligned} \tag{14}$$

where the symmetric matrix A has entries $A_{ii} = \left(\frac{L}{\hat{h}_i} \right)^2$ and $A_{ij} = -i \frac{L}{\hat{h}_{\min(i,j)}}$:

$$A = \begin{pmatrix} \frac{L^2}{\hat{h}_1^2} & -i \frac{L}{\hat{h}_1} & -i \frac{L}{\hat{h}_1} & \dots & -i \frac{L}{\hat{h}_1} \\ -i \frac{L}{\hat{h}_1} & \frac{L^2}{\hat{h}_2^2} & -i \frac{L}{\hat{h}_2} & \dots & -i \frac{L}{\hat{h}_2} \\ -i \frac{L}{\hat{h}_1} & -i \frac{L}{\hat{h}_2} & \frac{L^2}{\hat{h}_3^2} & \dots & -i \frac{L}{\hat{h}_3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \end{pmatrix}. \tag{15}$$

We can rewrite $A = D(I + K)D$ where D is a diagonal matrix whose entries are $D_{ii} = \frac{L}{\hat{h}_i}$, and K is a matrix with entries $K_{ij} = -i \frac{\hat{h}_{\max(i,j)}}{L}$ and 0 on the diagonal. Then $\det A = \det D^2 \det(I + K) = \frac{L^{2L}}{\prod_i \hat{h}_i^2} \det(I + K)$. Indeed $A_{ij} = D_{ii}(\delta_{ij} + K_{ij})D_{jj}$: for $i = j$ we get $A_{ii} = D_{ii}^2 = \frac{L^2}{\hat{h}_i^2}$, while for $i \neq j$ we get $A_{ij} = D_{ii}K_{ij}D_{jj} = \frac{L}{\hat{h}_i} \left(-i \frac{\hat{h}_{\max(i,j)}}{L} \right) \frac{L}{\hat{h}_j} = -i \frac{L}{\hat{h}_{\min(i,j)}}$ as required.

Putting everything together:

$$\int [dm d\hat{m}] e^{\sum_{\mu,i} \log \cos(\hat{h}_i m_i^\mu - \hat{M}_i^\mu) + iL \sum_{\mu,i} \hat{m}_i^\mu m_i^\mu} = \prod_\mu \frac{1}{\sqrt{\det(I + K)}} = e^{-\frac{M}{2} \log \det(I + K)} \tag{16}$$

Notice that elements of K are $O(\frac{1}{L})$ but they can still contribute to the determinant at leading order because K is dense. We expand $\log \det(I + K) = \text{Tr } K - \frac{1}{2} \text{Tr } K^2 + \frac{1}{3} \text{Tr } K^3 + \dots$. The first term is $\text{Tr } K = \sum_i K_{ii} = 0$ since K has zero diagonal. For the second term, we compute $\text{Tr } K^2 = \sum_{i,k} K_{ik} K_{ki} = -\frac{1}{L^2} \sum_i \sum_{k \neq i} \hat{h}_{\max(i,k)}^2 = -\frac{1}{L^2} \left[\sum_i \sum_{k<i} \hat{h}_{\max(i,k)}^2 + \sum_i \sum_{k>i} \hat{h}_{\max(i,k)}^2 \right] = -\frac{1}{L^2} \left[\sum_i \sum_{k<i} \hat{h}_i^2 + \sum_i \sum_{k>i} \hat{h}_k^2 \right] = -\frac{1}{L^2} \left[\sum_i \sum_{k<i} \hat{h}_i^2 + \sum_i \sum_{k>i} \hat{h}_k^2 \right] = -\frac{2}{L^2} \sum_i (i-1) \hat{h}_i^2$ hence:

$$\mathbb{E}_W Z_W(\psi) = \int [dh d\hat{h}] e^{\sum_i \log 2 \cosh(\psi_i) + \sum_i \log \sigma(\beta h_i) - i \sum_i \hat{h}_i h_i - \frac{\alpha}{2} \sum_i \frac{i-1}{L} \hat{h}_i^2} \tag{17}$$

We can factorise i and integrate $\hat{h}_i$:

$$\begin{aligned}
\mathbb{E}_W Z_W(\psi) &= \prod_i \int \frac{dh_i}{\sqrt{2\pi}} \frac{1}{\sqrt{\alpha \frac{i-1}{L}}} e^{\log 2 \cosh(\psi_i) + \log \sigma(\beta h_i) - \frac{1}{2} \frac{h_i^2}{\alpha \frac{i-1}{L}}} \\
&= \prod_i \int Dh_i 2 \cosh(\psi_i) \sigma(\beta \sqrt{\alpha \frac{(i-1)}{L}} h_i) \\
&= \int \mathcal{D}h \exp \left(L \int_0^1 dt \log 2 \cosh(\psi(t)) + \log \sigma(\beta \sqrt{\alpha t} h(t)) \right)
\end{aligned} \tag{18}$$

Where in the last step we made explicit the continuous limit where we introduced the time $t = i/L$.

However if we were to include all terms, we would have to compute the determinant of $I + K$ exactly, which is not trivial. The recursive structure of A makes it however doable: define $c_i = -i \frac{\hat{h}_i}{L}$. Then:

$$(I + K)_L = \begin{pmatrix} (I + K)_{L-1} & c_L \mathbf{1} \\ c_L \mathbf{1}^\top & 1 \end{pmatrix}. \quad (19)$$

We can use Shur's formula:

$$\det(I + K)_L = \det(I + K)_{L-1} (1 - c_L^2 \mathbf{1}^\top (I + K)_{L-1}^{-1} \mathbf{1}). \quad (20)$$

Defining $q_L = \mathbf{1}^\top (I + K)_L^{-1} \mathbf{1}$ one gets that $\det(I + K)_L = \prod_{i=2}^L (1 - c_i^2 q_{i-1})$. So once we have all L therms q_L , we have an expression for the determinant. This can be done again recursively: we can compute $(I + K)_L^{-1}$ via inversion of a block matrix:

$$(I + K)_L^{-1} = \begin{pmatrix} (I + K)_{L-1}^{-1} + \frac{c_L^2 (I + K)_{L-1}^{-1} \mathbf{1} \mathbf{1}^\top (I + K)_{L-1}^{-1}}{1 - c_L^2 q_{L-1}} & -\frac{c_L (I + K)_{L-1}^{-1} \mathbf{1}}{1 - c_L^2 q_{L-1}} \\ -\frac{c_L \mathbf{1}^\top (I + K)_{L-1}^{-1}}{1 - c_L^2 q_{L-1}} & \frac{1}{1 - c_L^2 q_{L-1}} \end{pmatrix} \quad (21)$$

and

$$\begin{aligned} q_L &= \mathbf{1}^\top (I + K)_L^{-1} \mathbf{1} = \mathbf{1}^\top \begin{pmatrix} (I + K)_{L-1}^{-1} \mathbf{1} + \frac{c_L^2 (I + K)_{L-1}^{-1} \mathbf{1} \mathbf{1}^\top (I + K)_{L-1}^{-1} \mathbf{1}}{1 - c_L^2 q_{L-1}} - \frac{c_L (I + K)_{L-1}^{-1} \mathbf{1}}{1 - c_L^2 q_{L-1}} \\ -\frac{c_L \mathbf{1}^\top (I + K)_{L-1}^{-1}}{1 - c_L^2 q_{L-1}} + \frac{1}{1 - c_L^2 q_{L-1}} \end{pmatrix} \\ &= q_{L-1} + \frac{c_L^2 q_{L-1}^2}{1 - c_L^2 q_{L-1}} - \frac{c_L q_{L-1}}{1 - c_L^2 q_{L-1}} - \frac{c_L q_{L-1}}{1 - c_L^2 q_{L-1}} + \frac{1}{1 - c_L^2 q_{L-1}} \\ &= q_{L-1} + \frac{(1 - c_L q_{L-1})^2}{1 - c_L^2 q_{L-1}}. \end{aligned} \quad (22)$$

Substituting the definition of c_i we get:

$$\log \det(I + K) = \sum_{i=2}^L \log \left(1 + \frac{\hat{h}_i^2}{L^2} q_{i-1} \right) \quad (23)$$

with of course $q_1 = 1$ and

$$q_L - q_{L-1} = \frac{\left(1 + \frac{i}{L} \hat{h}_L q_{L-1}\right)^2}{1 + \frac{\hat{h}_L^2}{L^2} q_{L-1}}. \quad (24)$$

We now go to the continuous limit introducing the time $t = i/L$. It must be that q_L must scale like L : indeed, putting $K = 0$, we have $q_L = \mathbf{1}^\top I \mathbf{1} = \sum_i i = L$. Parametrise $q_L = Lr(t)$, then the recursion becomes:

$$L(r(t + 1/L) - r(t)) = \frac{\left(1 + i \hat{h}(t) r(t)\right)^2}{1 + \frac{\hat{h}(t)^2}{L} r(t)} \quad (25)$$

the second term in the denominator vanishes, and we recognise the right hand side as a derivative:

$$\frac{dr}{dt} = \left(1 + i \hat{h}(t) r(t)\right)^2 \quad (26)$$

with initial condition $r(0) = 0$. In the continuous limit, the determinant becomes

$$\log \det(I + K) = \int_0^1 dt \hat{h}(t)^2 r(t). \quad (27)$$

The full expression for the generating functional in the continuous limit is then

$$\begin{aligned}\mathbb{E}_W Z_W(\psi) &= \int \prod_i \frac{dh_i d\hat{h}_i}{2\pi} e^{\sum_i \log 2 \cosh(\psi_i) + \sum_i \log \sigma(\beta h_i) - i \sum_i \hat{h}_i h_i - \frac{M}{2} \log \det(I+K)} \\ &= \int \mathcal{D}h \mathcal{D}\hat{h} e^{LS[h, \hat{h}]}\end{aligned}\tag{28}$$

with $S[h, \hat{h}] = \int_0^1 dt s(t)$ and

$$s = \log 2 \cosh(\psi) + \log \sigma(\beta h) - i \hat{h} h - \frac{\alpha}{2} \hat{h}^2 r + \lambda \left(\dot{r} - (1 + i \hat{h} r)^2 \right)\tag{29}$$

where we enforced the definition of $r(t)$ via a Lagrange multiplier $\lambda(t)$ in the action.

However, both the gaussian closure and the exact dynamical equations are trivially useless in absence of a signal to be retrieved. We therefore need to look for solutions with a macroscopic overlap with one of the patterns, which will be the retrieved one. This is the subject of the next section.

1.3 Condensed regime

To capture pattern retrieval we single out one pattern, say ξ^1 , and look for solutions with macroscopic overlap

$$m_i := m_i^{\mu=1} = \frac{1}{L} \sum_{j < i} \xi_j^{\mu=1} x_j = O(1),\tag{30}$$

while the remaining $M - 1$ patterns stay in the paramagnetic regime $m_i^\mu = O\left(\frac{1}{\sqrt{L}}\right)$ for $\mu > 1$. Here we don't have to average on ξ^1 since the retrieval performance will be read off by conditioning on a specific value of ξ^1 . For this reason, we introduce the spin variable $s_i = x_i \xi_i^{\mu=1}$.

$$\begin{aligned}\mathbb{E}_W Z_W(\psi | \xi^{\mu=1}) &= \mathbb{E}_{\{\xi^\mu\}_{\mu>1}} \int [dh d\hat{h}] \sum_x e^{\sum_i \psi_i x_i - i \sum_i \hat{h}_i h_i + i \frac{1}{L} \sum_i \hat{h}_i x_i \sum_{j < i} \xi_i^{\mu=1} \xi_j^{\mu=1} x_j} \times \\ &\times e^{i \frac{1}{L} \sum_{\mu>1, i} \hat{h}_i x_i \sum_{j < i} \xi_i^\mu \xi_j^\mu x_j} \prod_i \sigma(\beta h_i) = \\ &= \mathbb{E}_{\{\xi^\mu\}_{\mu>1}} \int [dh d\hat{h}] \sum_s e^{\sum_i \psi_i \xi_i^{\mu=1} s_i - i \sum_i \hat{h}_i h_i + i \frac{1}{L} \sum_i \hat{h}_i s_i \sum_{j < i} s_j} \times \\ &\times e^{i \frac{1}{L} \sum_{\mu>1, i} \hat{h}_i s_i \xi_i^{\mu=1} \xi_i^\mu \sum_{j < i} \xi_j^\mu \xi_j^{\mu=1} s_j} \prod_i \sigma(\beta h_i)\end{aligned}\tag{31}$$

One can define new patterns $\tilde{\xi}_i^\mu := \xi_i^{\mu=1} \xi_i^\mu$ which are again iid Rademacher ± 1 , so the disorder average factorises across patterns as before and we get the same result (with a vanishing difference due to $M - 1$). Notice that it goes out of the sum over $\{s\}$ due to the cosine trick of before. We can also harmlessly absorb $\xi^{\mu=1}$ into ψ as it is always taken in its 0 limit when used. For the macroscopic magnetisation term, we enforce $m_{i+1} = m_i + \frac{1}{L} s_i$:

$$\begin{aligned}\mathbb{E}_W Z_W(\psi | \xi^{\mu=1}) &= \int [dh d\hat{h}] \prod_i \frac{L dm_i d\hat{m}_i}{2\pi} \sum_s e^{\sum_i \psi_i s_i + i \sum_i \hat{h}_i s_i m_i - i \sum_i \hat{m}_i s_i} \times \\ &\times e^{iL \sum_i (m_{i+1} - m_i) - i \sum_i \hat{h}_i h_i - \frac{M-1}{2} \log \det(I+K[\hat{h}]) + \sum_i \log \sigma(\beta h_i)} = \\ &= \int [dh d\hat{h}] \prod_i \frac{L dm_i d\hat{m}_i}{2\pi} e^{\sum_i \log 2 \cosh(\psi_i + i \hat{h}_i m_i - i \hat{m}_i)} \times \\ &\times e^{iL \sum_i (m_{i+1} - m_i) - i \sum_i \hat{h}_i h_i - \frac{M-1}{2} \log \det(I+K[\hat{h}]) + \sum_i \log \sigma(\beta h_i)}.\end{aligned}\tag{32}$$

In the continuous limit this is:

$$\mathbb{E}_W Z_W(\psi|\xi^{\mu=1}) = \int \mathcal{D}h \mathcal{D}\hat{h} \mathcal{D}m \mathcal{D}\hat{m} \mathcal{D}\lambda \mathcal{D}r e^{LS[h, \hat{h}, m, \hat{m}, \lambda, r]} \quad (33)$$

With the action $S = \int_0^1 dt s(t)$ and

$$\begin{aligned} s = & \log \sigma(\beta h) + \log 2 \cosh(\psi + i\hat{h}m - i\hat{m}) + \\ & + i\hat{m}\dot{m} - i\hat{h}h - \frac{\alpha}{2}\hat{h}^2r + \lambda(\dot{r} - (1 + i\hat{h}r)^2). \end{aligned} \quad (34)$$

It would be tempting to do a functional saddle point on the whole functional, but one must not forget that the field h is a fluctuating quantity even at $L \rightarrow \infty$: as we saw in the pedagogical paramagnetic closure it was gaussian with variance $\alpha t = O(1)$. We must therefore integrate the h and $\hat{h}$ fields first.

$$\begin{aligned} s = & \log \sigma(\beta h) + \log 2 \cosh(\psi + i\hat{h}m - i\hat{m}) + \\ & + i\hat{m}\dot{m} - i\hat{h}h - \frac{\alpha}{2}\hat{h}^2r + \lambda(\dot{r} - 1) + \lambda\hat{h}^2r^2 - 2i\lambda\hat{h}r = \\ = & -\frac{1}{2}(\alpha r - 2\lambda r^2)\hat{h}^2 - (ih + 2i\lambda r)\hat{h} + \log 2 \cosh(\psi + i\hat{h}m - i\hat{m}) + \\ & + i\hat{m}\dot{m} + \lambda(\dot{r} - 1) + \log \sigma(\beta h). \end{aligned} \quad (35)$$

Due to the exponential nature of the integral, the gaussian integrals can be performed at fixed time t and then put back together in the continuous limit. The first one is

$$\begin{aligned} \int \frac{d\hat{h}}{\sqrt{2\pi}} e^{-\frac{1}{2}(\alpha r - 2\lambda r^2)\hat{h}^2 - (2i\lambda r + ih)\hat{h} + \log 2 \cosh(\psi + i\hat{h}m - i\hat{m})} = \\ = \int \frac{d\hat{h}}{\sqrt{2\pi}} e^{-\frac{1}{2}(\alpha r - 2\lambda r^2)\hat{h}^2 - (2i\lambda r + ih)\hat{h}} \left[e^{\psi + i\hat{h}m - i\hat{m}} + e^{-(\psi + i\hat{h}m - i\hat{m})} \right] \end{aligned} \quad (36)$$

We have a sum of the two integrals of the form

$$\begin{aligned} \int \frac{d\hat{h}}{\sqrt{2\pi}} e^{-\frac{1}{2}(\alpha r - 2\lambda r^2)\hat{h}^2 + (2i\lambda r - ih)\hat{h}} \left[e^{\pm(\psi + i\hat{h}m - i\hat{m})} \right] = \\ = \frac{1}{\sqrt{\alpha r - 2\lambda r^2}} e^{\pm(\psi - i\hat{m}) + \frac{(2\lambda r + h \pm m)^2}{2(\alpha r - 2\lambda r^2)}}. \end{aligned} \quad (37)$$

Then:

$$\begin{aligned} \frac{1}{\sqrt{\alpha r - 2\lambda r^2}} \int \frac{dh}{\sqrt{2\pi}} \sigma(\beta h) e^{\frac{(2\lambda r + h \pm m)^2}{2(\alpha r - 2\lambda r^2)}} = \\ = \int Dz \sigma(\beta(\pm m + 2\lambda r + \sqrt{\alpha r - 2\lambda r^2}z)) \end{aligned} \quad (38)$$

After doing the last integral, we put everything back in the continuous limit and obtain the generating functional action (from now on we put $\psi = 0$ because I am pretty sure it is useless)

$$\begin{aligned} s = & i\hat{m}\dot{m} + \lambda(\dot{r} - 1) + \log \sum_{p=\pm 1} e^{-ip\hat{m}} \int Dz \sigma(\beta(pm + 2\lambda r + \sqrt{\alpha r - 2\lambda r^2}z)) \\ = & i\hat{m}\dot{m} + \lambda(\dot{r} - 1) + \log[G_+(m, \hat{m}, r, \lambda) + G_-(m, \hat{m}, r, \lambda)]. \end{aligned} \quad (39)$$

The symmetry of σ lets Euler Lagrange equations for $\hat{m}$ and λ give the symmetric $\hat{m} = 0$ and $\lambda = 0$ as solutions. Indeed, with $G_{\pm}$ defined as above, one has:

$$\begin{aligned} G_+(m, \hat{m} = 0, r, \lambda = 0) &= 1 - G_-(m, \hat{m} = 0, r, \lambda = 0) \\ \partial_m G(m, \hat{m} = 0, r, \lambda = 0) &= -\partial_m G(m, \hat{m} = 0, r, \lambda = 0) \\ \partial_r G(m, \hat{m} = 0, r, \lambda = 0) &= -\partial_r G(m, \hat{m} = 0, r, \lambda = 0) \end{aligned} \quad (40)$$

Then the Euler-Lagrange equations on $\hat{m}$ and $\dot{\lambda}$ give:

$$i\dot{\hat{m}} = \frac{\partial_m G_+(m, \hat{m}, r, \lambda) + \partial_m G_-(m, \hat{m}, r, \lambda)}{G_+(m, \hat{m}, r, \lambda) + G_-(m, \hat{m}, r, \lambda)} = 0 \quad (41)$$

and

$$\dot{\lambda} = \frac{\partial_r G_+(m, \hat{m}, r, \lambda) + \partial_r G_-(m, \hat{m}, r, \lambda)}{G_+(m, \hat{m}, r, \lambda) + G_-(m, \hat{m}, r, \lambda)} = 0 \quad (42)$$

which are satisfied by $\hat{m} = 0$ and $\lambda = 0$. Instead, differentiating with respect to $\hat{m}$ and inserting the symmetric soltion we get the familiar

$$\dot{m} = 1 - 2 \int Dz \sigma(\beta(m + \sqrt{\alpha r} z)) \quad (43)$$

We just have to find the equation for r : varying with respect to λ we get:

$$\dot{r} = 1 - \frac{\partial_\lambda G_+(m, \hat{m}, r, \lambda) + \partial_\lambda G_-(m, \hat{m}, r, \lambda)}{G_+(m, \hat{m}, r, \lambda) + G_-(m, \hat{m}, r, \lambda)} \quad (44)$$

since

$$\partial_\lambda G_\pm(m, \hat{m}, r, \lambda) = \beta e^{\mp i \hat{m}} \int Dz \sigma'(\beta(\pm m + 2\lambda r + \sqrt{\alpha r - 2\lambda r^2} z)) \left[2r - \frac{r^2}{\sqrt{\alpha r - 2\lambda r^2}} z \right] \quad (45)$$

we have

$$\begin{aligned} & \partial_\lambda G_+(m, \hat{m} = 0, r, \lambda = 0) + \partial_\lambda G_-(m, \hat{m} = 0, r, \lambda = 0) = \\ & = \beta \int Dz [\sigma'(\beta(m + \sqrt{\alpha r} z)) + \sigma'(\beta(-m + \sqrt{\alpha r} z))] \left[2r - \frac{r^2}{\sqrt{\alpha r}} z \right] \\ & = 4\beta r \int Dz \sigma'(\beta(m + \sqrt{\alpha r} z)) \end{aligned} \quad (46)$$

where we used the symmetry of σ to get rid of the second term and double the first one. The complete equations are then:

$$\begin{aligned} \dot{m} &= 2 \int Dz \sigma(\beta(m + \sqrt{v} z)) - 1 \\ \dot{v} &= \alpha - 4\beta v \int Dz \sigma'(\beta(m + \sqrt{v} z)). \end{aligned} \quad (47)$$

Where we defined $v = \alpha r$. We discover that the equation of our previous work is obtained by neglecting this last susceptibility term $4\beta v \int Dz \sigma'(\beta(m + \sqrt{v} z))$. Specialising for the to algorithms “fair” ($\beta = 1$) and “greedy” ($\beta = \infty$) we get:

$$\begin{aligned} \dot{m} &= 2 \int Dz \sigma(m + \sqrt{v} z) - 1 \\ \dot{v} &= \alpha - 4v \int Dz \sigma'(m + \sqrt{v} z) \end{aligned} \quad (48)$$

and

$$\begin{aligned} \dot{m} &= 2\Phi\left(\frac{m}{\sqrt{v}}\right) - 1 \\ \dot{v} &= \alpha - 4\sqrt{v}\varphi\left(\frac{m}{\sqrt{v}}\right). \end{aligned} \quad (49)$$

2 Autoregressive network with learned weights

Suppose now the weights have been obtained by minimising the following loss:

$$\mathcal{L}_i(\{w_{ij}\}_{j=1}^L, \{\xi^\mu\}_{\mu=1}^M) = -\frac{1}{M} \sum_{\mu=1}^M \mathbb{E}_t \frac{1}{t} \mathbb{E}_{m^t} \mathbb{I}(m_i^t = 1) \log \sigma \left(\frac{1}{\sqrt{L}} \xi_i^\mu \sum_j (w_j (1 - m_j^t) \xi_j^\mu) \right) \quad (50)$$

the MSR generating functional is then

$$\begin{aligned} \mathbb{E}_W Z_W(\psi) &= \mathbb{E}_W \int [dh d\hat{h}] \sum_x e^{-i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i x_i \sum_{j<i} \frac{W_{ij}}{\sqrt{L}} x_j + \sum_i \log \sigma(h_i)} \\ &= \mathbb{E}_{\{\xi^\mu\}_{\mu=1}^M} \int \prod_{ij} dw_{ij} e^{-\beta M \sum_i \mathcal{L}_i(\{w_{ij}\}_{j=1}^L, \{\xi^\mu\}_{\mu=1}^M) - \frac{1}{2} \beta \sum_{ij} \lambda \|w_{ij}\|^2} \times \\ &\quad \times \int [dh d\hat{h}] \sum_x e^{-i \sum_i \hat{h}_i h_i + i \frac{1}{L} \sum_{\mu,i} \hat{h}_i x_i \sum_{j<i} \frac{w_{ij}}{\sqrt{L}} x_j + \sum_i \log \sigma(h_i)} \end{aligned} \quad (51)$$

2.1 Gaussian Ansatz

A first simple approximation is to use our knowledge of W from the replica computation. Assume W_{ij} elements are Gaussian with row variance $\frac{1}{L} \sum_j W_{ij}^2 = q$ and row mean $\frac{1}{\sqrt{L}} \sum_j W_{ij} = \mu_i$ where for each row $\mu_i \sim \mu(x)$ where $x \sim \mathcal{N}(0, 1)$ iid. Then we select a pattern $\xi^{\mu=1}$ and define $s_i = \xi_i^{\mu=1} x_i$:

$$\begin{aligned} \mathbb{E}_W Z_W(\psi | \xi^{\mu=1}) &= \mathbb{E}_W \int [dh d\hat{h}] \sum_x e^{-i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i x_i \sum_{j<i} \frac{W_{ij}}{\sqrt{L}} x_j + \sum_i \log \sigma(h_i)} \\ &= \mathbb{E}_W \int [dh d\hat{h}] \sum_s e^{-i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i s_i \sum_{j<i} \frac{J_{ij}}{\sqrt{L}} s_j + \sum_i \log \sigma(h_i)} \end{aligned} \quad (52)$$

where we defined $J_{ij} = x_i^{\mu=1} W_{ij} \xi_j^{\mu=1}$. The important observation is that J_{ij} are again Gaussian with row variance q and row mean μ_i for each row. We thus write: $J_{ij} = \frac{\mu_i}{\sqrt{L}} + \sqrt{q} z_{ij}$, where the z_{ij} are gaussian with the extra condition that $\sum_j z_{ij} = 0$. Notice that this enforces the gaussian vectors to live in the $L-1$ space orthogonal to $\mathbf{1}$, and the covariance matrix of each row of $z_i = (z_{i1}, z_{i2}, \dots)$ is $\Sigma_i = \mathbb{I} - \frac{1}{L} \mathbf{1}\mathbf{1}^\top$.

$$\begin{aligned} \mathbb{E}_W Z_W(\psi | \xi^{\mu=1}) &= \mathbb{E}_W \int [dh d\hat{h}] \sum_s e^{-i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i s_i \sum_{j<i} \frac{J_{ij}}{\sqrt{L}} s_j + \sum_i \log \sigma(h_i)} \\ &= \mathbb{E}_{\mu, z} \int [dh d\hat{h}] \sum_s e^{-i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i s_i \sum_{j<i} \frac{\frac{\mu_i}{\sqrt{L}} + \sqrt{q} z_{ij}}{\sqrt{L}} s_j + \sum_i \log \sigma(h_i)} \\ &= \mathbb{E}_{\mu, z} \int [dh d\hat{h}] \sum_s e^{-i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i s_i \mu_i \sum_{j<i} \frac{s_j}{L} + i \sum_i \hat{h}_i s_i \sqrt{q} \sum_{j<i} \frac{z_{ij}}{\sqrt{L}} s_j + \sum_i \log \sigma(h_i)} \\ &= \mathbb{E}_{\mu, z} \int [dh d\hat{h}] \sum_s e^{-i \sum_i \hat{h}_i h_i + i \sum_i \hat{h}_i s_i \mu_i m_i + i \sum_i \hat{h}_i s_i \sqrt{q} \sum_{j<i} \frac{z_{ij}}{\sqrt{L}} s_j + \sum_i \log \sigma(h_i)} \end{aligned} \quad (53)$$

We do the first Gaussian integral on the z_{ij} factorizing by row i :

$$\int \prod_j D z_{ij} e^{i \sum_i \hat{h}_i s_i \sqrt{q} \sum_{j<i} \frac{z_{ij}}{\sqrt{L}} s_j} = e^{\frac{1}{2} a^\top \Sigma_i a} \quad (54)$$

where $a_j = i \hat{h}_i \sqrt{\frac{q}{L}} s_i s_j \mathbf{1}_{j<i}$. We have

$$a^\top \Sigma_i a = a^\top a - \frac{1}{L} (a^\top \mathbf{1})^2 = -q \hat{h}_i^2 \left[\frac{i-1}{L} - \left(\frac{1}{L} \sum_{j<i} s_j \right)^2 \right] \quad (55)$$

We now enforce $m_{i+1} = m_i + \frac{1}{L}s_i$: notice that it also imposes $\frac{1}{L}\sum_{j<i}s_j = m_i$ recursively.

$$\begin{aligned}
\mathbb{E}_W Z_W(\psi|\xi^{\mu=1}) &= \\
&= \mathbb{E}_\mu \int [dm d\hat{m}] [dh d\hat{h}] \sum_s e^{-i\sum_i \hat{h}_i h_i + i\sum_i \hat{h}_i s_i \mu_i m_i - \frac{1}{2}q \sum_i [\frac{i-1}{L} - m_i^2] \hat{h}_i^2 + \sum_i \log \sigma(h_i) + iL \sum_i \hat{m}_i (m_{i+1} - m_i - \frac{1}{L}s_i)} \\
&= \mathbb{E}_\mu \int [dm d\hat{m}] [dh d\hat{h}] \sum_s \prod_i e^{-i\hat{h}_i h_i + i\hat{h}_i s_i \mu_i m_i - \frac{1}{2}q [\frac{i-1}{L} - m_i^2] \hat{h}_i^2 + \log \sigma(h_i) + iL \hat{m}_i (m_{i+1} - m_i - \frac{1}{L}s_i)} \\
&= \mathbb{E}_\mu \int [dm d\hat{m}] [dh d\hat{h}] \prod_i e^{-i\hat{h}_i h_i - \frac{1}{2}q [\frac{i-1}{L} - m_i^2] \hat{h}_i^2 + \log \sigma(h_i) + iL \hat{m}_i (m_{i+1} - m_i)} \sum_{s_i=\pm 1} e^{i\hat{h}_i s_i \mu_i m_i - i\hat{m}_i s_i} \\
&= \mathbb{E}_\mu \int [dm d\hat{m}] [dh d\hat{h}] \prod_i e^{-i\hat{h}_i h_i - \frac{1}{2}q [\frac{i-1}{L} - m_i^2] \hat{h}_i^2 + \log \sigma(h_i) + iL \hat{m}_i (m_{i+1} - m_i) + \log 2 \cosh i(\hat{h}_i \mu_i m_i - \hat{m}_i)}
\end{aligned} \tag{56}$$

performing the integrals in h and $\hat{h}$ as before we get, in the continuous limit:

$$\mathbb{E}_W Z_W(\psi|\xi^{\mu=1}) = \mathbb{E}_\mu \int \mathcal{D}m \mathcal{D}\hat{m} e^{LS[m, \hat{m}]} \tag{57}$$

where s such that $S[m, \hat{m}] = \int_0^1 dt s(t)$ is

$$s = i\hat{m}\dot{m} + \log \sum_{p=\pm 1} e^{-ip\hat{m}} \int Dz Dx \sigma(\beta(p\mu(x)m + \sqrt{q(t-m^2)}z)) \tag{58}$$

Clearly the fixed points solution is, as we found in the other metod:

$$\dot{m} = 2 \int Dz Dx \sigma(\beta(\mu(x)m + \sqrt{q(t-m^2)}z)) - 1. \tag{59}$$

2.2 Hebbianity of learned weights

We notice that this description completely misses the susceptibility term of the Hebbian case. It is then natural to start look at to what order the learned matrix W is close to the Hebbian one.

Define $H_{ij} = \frac{1}{\sqrt{L}} \sum_\mu \xi_i^\mu \xi_j^\mu$. We look at the rows $H_i = (H_{i1}, \dots, H_{iL})$ and $W_i = (W_{i1}, \dots, W_{iL})$ as vectors in L -dimensional space, and assume each row is independent from each other. We can then decompose:

$$W_i = \frac{W_i \cdot H_i}{\|H_i\|^2} H_i + R_i. \tag{60}$$

We already know $\frac{1}{L}\|H_i\|^2 = \frac{1}{L^2} \sum_j \sum_{\mu, \nu} \xi_i^\mu \xi_j^\nu \xi_i^\nu \xi_j^\mu = \alpha$. Then the projection of W_i on H_i is

$$W_i \cdot H_i = \frac{1}{L} \sum_j W_{ij} \left(\sum_\mu \frac{1}{\sqrt{L}} \xi_i^\mu \xi_j^\mu \right) = \frac{\alpha}{M} \sum_\mu \left(\frac{1}{\sqrt{L}} \sum_j W_{ij} \xi_i^\mu \xi_j^\mu \right) \tag{61}$$

By the usual properties of the row mean, each element $\left(\frac{1}{\sqrt{L}} \sum_j W_{ij} \xi_i^\mu \xi_j^\mu \right)$ of the sum over the pattens is again some $\mu \sim \mu(x)$ with $x \sim \mathcal{N}(0, 1)$. The empirical average over all infinite patterns makes this term converge to $\mathbb{E}\mu(x)$. Therefore, the projection of W_i on H_i is equal for all rows and it is

$$\gamma = \frac{W_i \cdot H_i}{\|H_i\|^2} = \frac{\alpha \mathbb{E}\mu(x)}{\alpha} = \int Dx \mu(x). \tag{62}$$

The cosine similarity between W_i and H_i is then

$$\cos(\theta_i) = \frac{W_i \cdot H_i}{\|W_i\| \|H_i\|} = \gamma \sqrt{\frac{\alpha}{q}}. \tag{63}$$

As we see from plots, it gets indeed very close to 1 for both regularisation values.

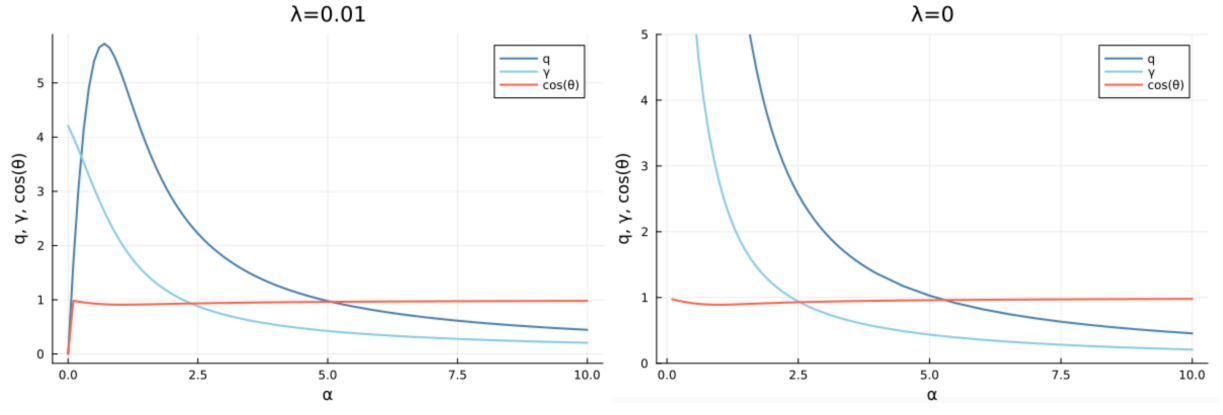


Figure 1: value of γ for the two regularisation values, and cosine similarity between W_i and H_i for each row.

APPENDIX A Standard Hopfield Setting

In Standard Hopfield I would have an annealed partition function. Setting $y_i^\mu := x_i \xi_i^\mu$, $(y_i^\mu)^2 = 1$, so

$$\sum_i x_i \sum_{j < i} \xi_i^\mu \xi_j^\mu x_j = \sum_{i > j} y_i^\mu y_j^\mu = \frac{1}{2} \left[\left(\sum_i y_i^\mu \right)^2 - \sum_i (y_i^\mu)^2 \right] = \frac{1}{2} \left[\left(\sum_i x_i \xi_i^\mu \right)^2 - L \right]. \quad (64)$$

Therefore

$$\begin{aligned} \mathbb{E}_\xi \sum_x e^{\frac{1}{L} \sum_\mu \sum_i x_i \sum_{j < i} \xi_i^\mu \xi_j^\mu x_j} &= e^{-\frac{M}{2}} \mathbb{E}_\xi \sum_x e^{\frac{1}{2L} \sum_\mu \left(\sum_i x_i \xi_i^\mu \right)^2} \\ &= e^{-\frac{M}{2}} \mathbb{E}_\xi \int \prod_\mu dm_\mu e^{-\frac{L}{2} \sum_\mu m_\mu^2} \sum_x e^{\sum_\mu m_\mu \sum_i x_i \xi_i^\mu}, \end{aligned} \quad (65)$$

where the trajectory-independent prefactor $e^{-\frac{M}{2}} = e^{-\alpha \frac{L}{2}}$ comes from the diagonal $\sum_i (y_i^\mu)^2 = L$ piece. This factor cancels against an identical contribution coming from the diagonal $i = j$ piece of the standard $\sum_{i,j} W_{ij} x_i x_j$ Hopfield energy, and is therefore commonly dropped together with x -independent normalisation constants. Keeping it explicit is required if one wants to compare the autoregressive and symmetric annealed free energies directly.